Design and Evaluation of an Explainable Legal AI for Bangladeshi Traffic Law

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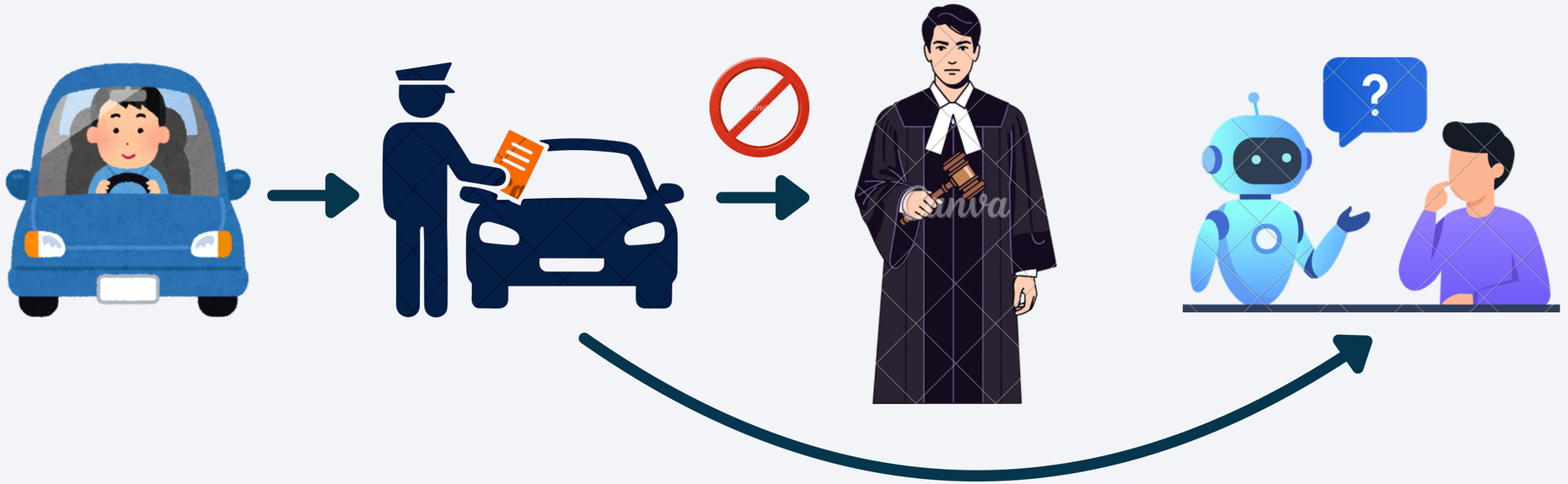
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Aminul Islam Rifat: 0112320100

Ifrat-E-Sarwar Orpa: 0112320141

Problem Statement



Driver driving car

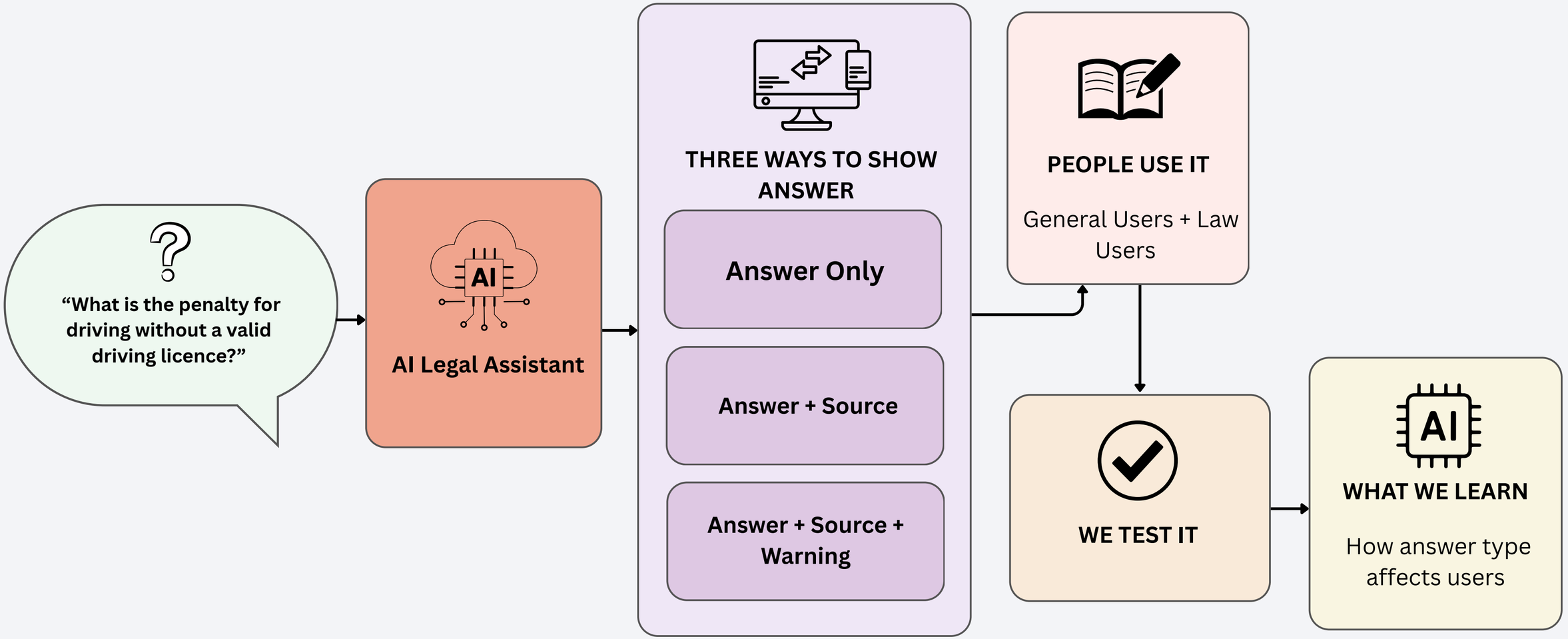
Gets a traffic violation ticket

Avoid a lawyer or legal person

Asks AI ChatBot instead

People increasingly use AI for legal advice, but its answers may not always be reliable.

Graphical Abstract



PROBLEM ARTICULATION

AI-generated Answer

↓

May contain an error

↓

User may not recognize the error

Core Problem

Difficulty in recognizing incorrect AI-generated legal information

OBJECTIVES

01 **Build a validated Bangla legal QA dataset.**

02 **Develop a RAG-based legal assistant.**

03 **Compare three interface designs.**

04 **Evaluate trust and usability.**

Motivation



Legal AI Proliferation



High Risk of Hallucinations



Unequal Ability to Catch Errors

Selected Focus :Road Transport Act(2018)

01 Legal AI Proliferation

AI legal assistants are becoming mainstream in Bangladesh and worldwide

- **Barriers to Access**
- **Public Turns to AI**
- **Rapid Mainstream Growth**



02 High Risk of Hallucination

Confident-sounding wrong answers are especially dangerous in law — and the risk grows in low-resource languages like Bangla

- **Fabricating Clauses**
- **Misinterpreting Sections**
- **Blending Outdated Laws**





03 Unequal Ability to Catch Errors

NOT EVERYONE CAN SPOT AN AI MISTAKE EQUALLY WELL — A "GOOD" INTERFACE FOR ONE USER CAN MISLEAD ANOTHER



Legal Expert

Can often recognize a wrong citation or a misapplied section on sight.



Everyday Citizen/Laymen

Usually cannot tell a fabricated clause from a real one — and may simply trust it.

This is why our study tests interface design across both user groups — good design cannot be one-size-fits-all.

Research Gaps & Novelty

Gap Area	Existing Work	Our Approach
LEGAL DOMAIN	**GENERAL LEGAL QA:** Broad law coverage	**TRAFFIC-LAW FOCUS:** Bangladesh Road Transport Act
EVALUATION	**SYSTEM-CENTRIC:** Accuracy and retrieval metrics	**HUMAN-CENTRIC:** Hallucination detection and trust
TRANSPARENCY	**LIMITED COMPARISON:** Few UI-level studies	**3-INTERFACE STUDY:** Baseline, Traceability, Explainable
USER GROUPS	**LIMITED COMPARISON:** Expertise often not studied	**EXPERTISE-AWARE:** Lay users vs. legal experts
DATASET	**LIMITED RESOURCE:** Native-Bangla traffic-law QA	**VALIDATED DATASET:** Expert-reviewed Bangla QA

OUR CONTRIBUTIONS

01 Evaluate hallucination detection and trust.

02 Compare three transparency interfaces.

03 Compare laymen and legal experts.

04 Provide a validated Bangla legal QA test set.

Literature Review

Work	Focus	Technique
JusticeNetBD (2025)	Legal accessibility for Bangladeshi women	Bangladesh-specific legal RAG
LegalRAG (2025)	Bangla-English legal QA	→ Multilingual retrieval → Query refinement
LawRAG(2025)	Legal retrieval & evaluation	→ Hybrid retrieval → Temporal/legal evaluation
Mina (2026)	Scalable legal assistance	→ Multilingual agentic RAG → Two-stage retrieval
JuristRAG (2025)	Complex legal queries & hallucination risk	→ Query decomposition → Iterative retrieval
Context-Injected Fine-Tuning for BD Law(2026)	LoRA on context-injected prompts	MCQ-only, no retrieval or traffic law
Exploring Possibilities of AI-Powered Legal Assistance (2024)	Bangladesh-specific legal AI assistant	→ Improve access to legal help in Bangladesh → Reduce problems caused by high cost, complexity and delays

Gap Finding

Model/Paper Name	Mina	JuristRAG	JusticeNetBD	LegalRAG	LawRAG	Proposed Model
BD Traffic Law	✗	✗	✗	✗	✗	✓
Bangla Focused	✓	✗	✗	✓	✗	✓
Legal-Structure Retrieval	✓	✓	✗	✗	◐	✓
Hybrid Retrieval	◐	✓	✗	✗	✓	✓
Temporal Awareness	✗	✗	✗	◐	✓	✓
Citation Verification	✗	✗	✗	✗	◐	✓
Retrieval Evaluation	✓	✓	✓	✗	✓	✓
Legal Expert Evaluation	✓	✓	✗	✓	✓	✓

Implemented = ✓ Not Implemented = ✗ Partially Implemented = ◐

Proposed Methodology

Data Foundation

01 Problem Definition & Scope

- Traffic-law legal assistant

02 Legal Data Collection

- Collect authoritative traffic-law sources
- Road Transport Act, 2018
- **Possible extension**: Road Transport Rules, 2022

03 Document Processing

- Clean and extract text
- Structure by section/clause
- Embedding Model
- Vector Databse

System Development

04 RAG-Based Legal Assistant Development

05 Three User Interface Variants

A. Baseline Chatbot
Answer only

B. Traceability Interface
Answer + legal citations

C. Explainable Interface
Answer + citations + evidence/uncertainty cues

User Study & Evaluation

06 Experimental Study Design

1. General Citizens
2. Legal User

Both groups interact with the three interface variants

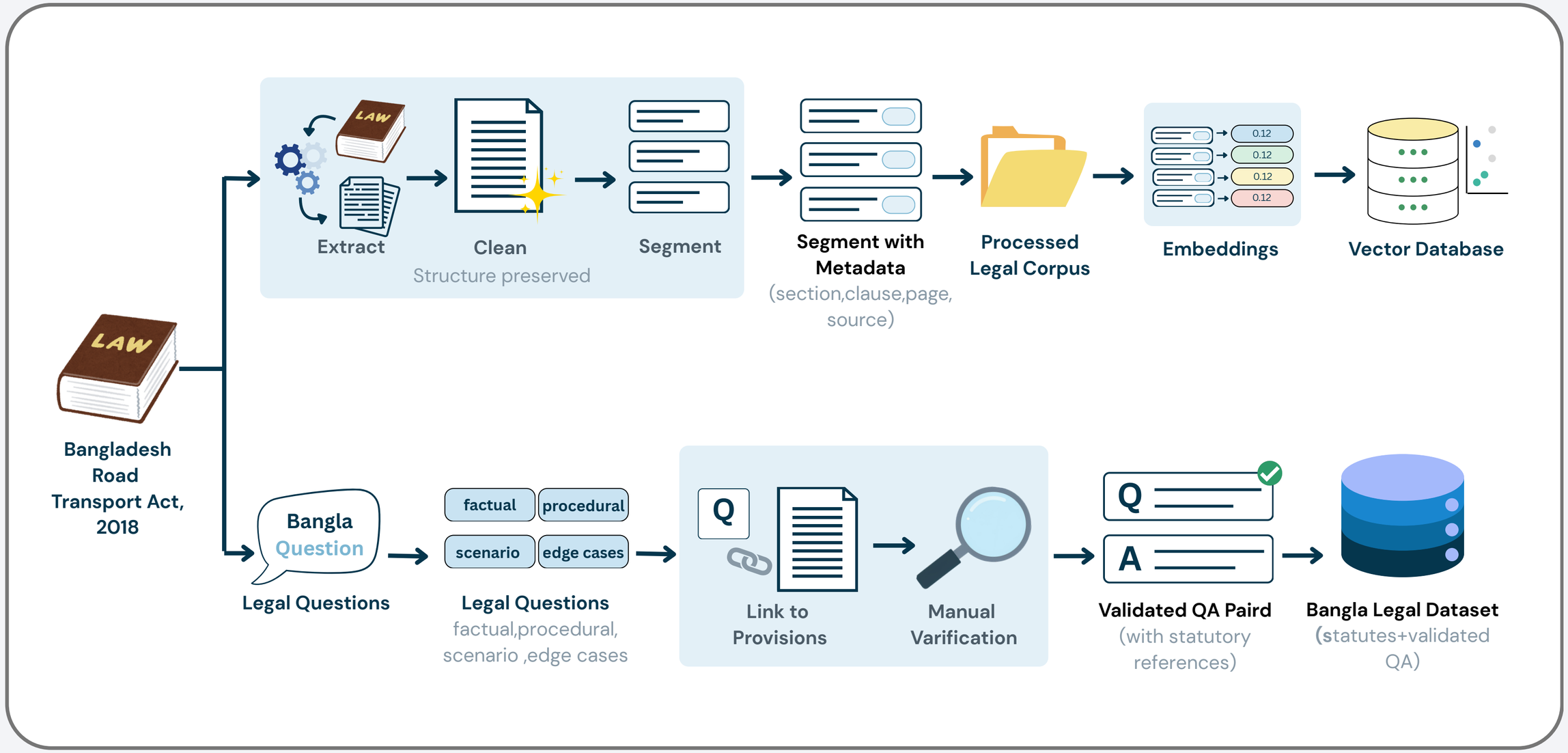
07 Evaluation Metrics

- Trust Calibration
- Hallucination Detection
- Cognitive Workload
- System Usability

08 Expected Outputs

- Explainable legal assistant prototype
- Structured Bangladesh traffic-law knowledge base
- Empirical user-study findings

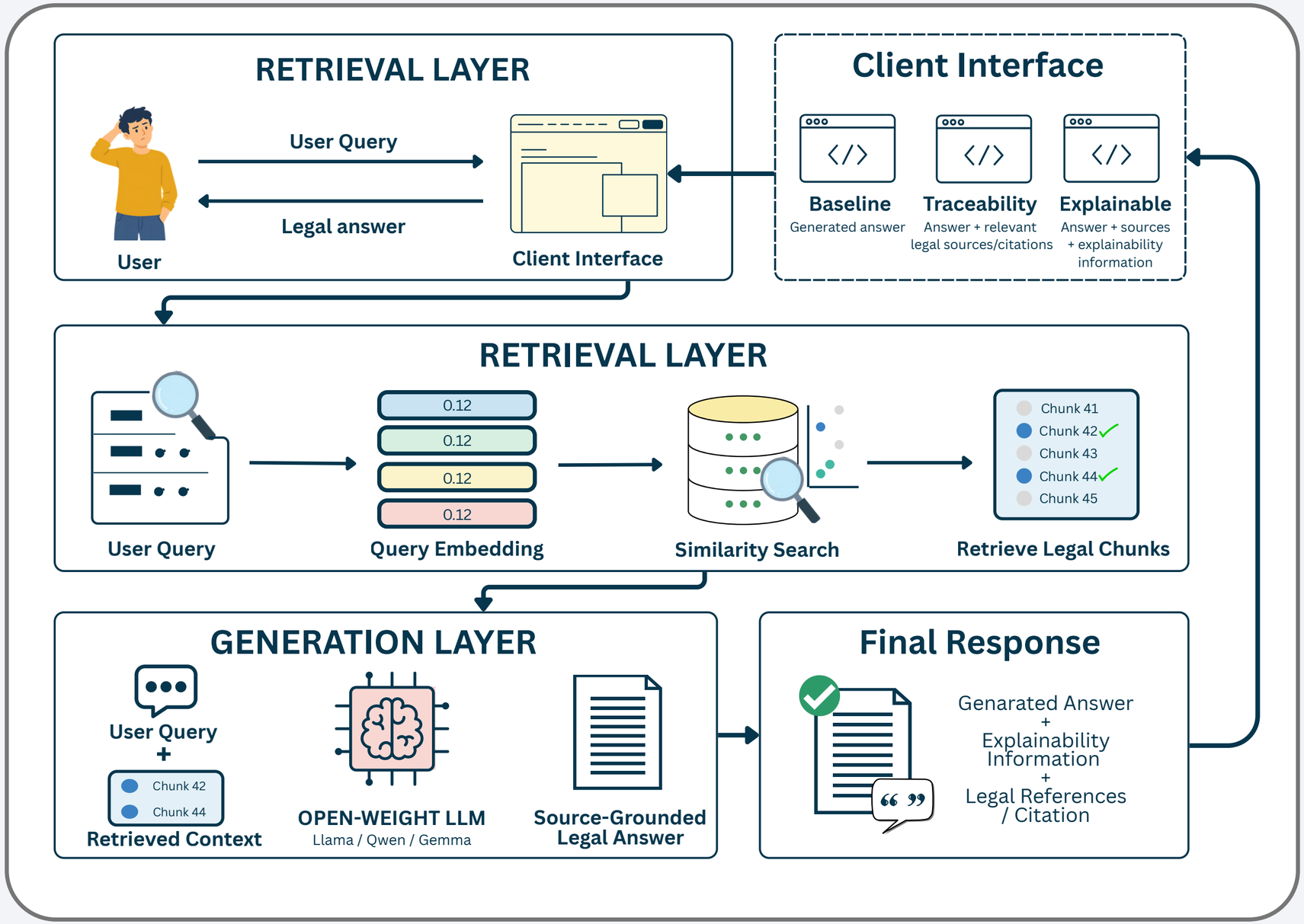
Data Pre-Processing



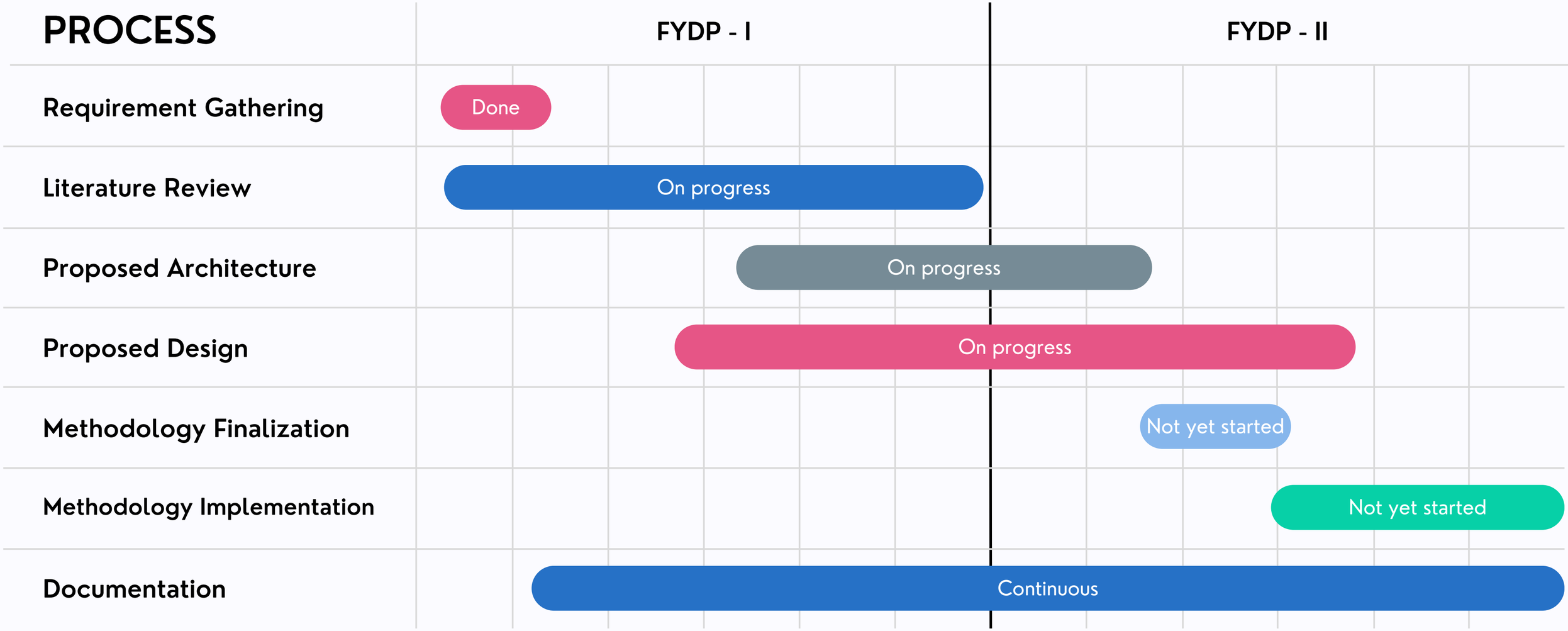
Example Evaluation Dataset

Question ID	Question Type	Question	Act	Section	Ground Truth	Label
Q001	Penalty	লাইসেন্স ছাড়া গাড়ি চালালে কী শাস্তি হতে পারে?	Road Transport Act, 2018	Sec. 38	Penalty defined by the cited provision	Supported
Q002	Procedure	ড্রাইভিং লাইসেন্স কীভাবে নবায়ন করা যায়?	Road Transport Act, 2018	Sec. 24	Renewal process as stated in the law	Supported
Q003	Scenario	একজন চালকের লাইসেন্সের মেয়াদ শেষ হলে কী হবে?	Road Transport Act, 2018	Sec. 24	Expired licence consequences from the provision	Partially Supported

Model Architecture



PROJECT SCHEDULE GANTT CHART



Application

01 Citizen Legal Tool.

02 Institutional Support.

03 Developer API.

04 Research, Education, and Design Guidelines.

Future Work

01 Broader Legal Coverage.

02 Multilingual & Code-Mixed Queries.

03 Stronger Reliability Mechanisms.

04 Large-Scale Real-World Evaluation.

Conclusion

01 Legal RAG needs safer human–AI interaction.

02 Built a legal RAG assistant for Bangladesh.

03 Compared three interface designs.

04 Evaluated hallucination detection and trust.

Alignment with BAETE Complex Engineering Problem

P1

In-Depth Knowledge

HCI, XAI, RAG, NLP,
Legal Information Retrieval,
Web Development, Statistics

P2

Conflicting Requirements

- Transparency vs. Usability
- Explainability vs. Cognitive Load
- Trust vs. Over-reliance

P3

Depth of Analysis

- Interface designs
- User testing
- Trust & hallucination analysis

P6

Stakeholders

- Laymen
- Legal Experts
- Researchers

P7

Interdependence

Legal Documents → RAG System → User Interface → User Testing → Data Analysis

Alignment with BAETE Complex Engineering Activities

A1

Resources

- Legal Documents
- AI Models
- Vector Database
- Web Technologies
- Legal Experts
- User Participants

A2

Interaction

Integrating AI Accuracy, Transparency, and Usability

A3

Innovation

Combines RAG, XAI, and HCI to improve trust in Legal AI

A4

Societal Impact

Reduces risks of AI hallucinations in legal information

A5

Familiarity

Adapts existing HCI principles to the Bangladeshi Legal AI context

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The End

PROBLEM BACKGROUND

Legal Information

- Difficult for ordinary users to access and understand
- Contains specific legal terms and sections

AI-Assisted Legal Information

- LLMs allow users to ask questions in natural language
- Makes legal information more accessible

Hallucination Risk

- Fabricated legal clauses
- Misinterpreted sections
- Outdated legal information

An AI-generated legal answer may sound correct even when it is not.

Literature Review

Context-Injected Fine-Tuning for BD Law(2026)

→ LoRA on context-injected prompts
→ MCQ-only, no retrieval or traffic law

JusticeNetBD (2025)

Legal accessibility for Bangladeshi women
→ Bangladesh-specific legal RAG

LegalRAG (2025)

Bangla-English legal QA
→ Multilingual retrieval
→ Query refinement

LawRAG

Legal retrieval & evaluation
→ Hybrid retrieval
→ Temporal/legal evaluation

JuristRAG (2025)

Complex legal queries & hallucination risk
→ Query decomposition
→ Iterative retrieval

Mina (2026)

Scalable legal assistance
→ Multilingual agentic RAG
→ Two-stage retrieval

Exploring Possibilities of AI-Powered Legal Assistance (2024)

Bangladesh-specific legal AI assistant
→ Improve access to legal help in Bangladesh
→ Reduce problems caused by high cost, complexity and delays

Can We Trust AI With Legal Advice?







Project Overview



Problem

We noticed that many students struggle with staying organized during final exams, leading to stress and missed deadlines.



Solution

We designed a digital planner system specifically for university students to help manage tasks, track deadlines, and reduce stress.



Why It Matters

By making organization intuitive and accessible, this tool supports academic success and improves student well-being.

Objective

01 Build a validated Bangla legal QA dataset.

02 Develop a RAG-based legal assistant.

03 Compare three interface designs.

04 Evaluate trust and usability.

Graphical Abstract

“What is the penalty for driving without a valid driving licence?”

AI Legal Assistant
User Question
↓
Finds Relevant Law
↓
Provides Law-Based Answer

THREE WAYS TO SHOW ANSWER

- Answer Only
- Answer+ Source
- Answer+ Source +Warning

PEOPLE USE IT

General Users + Law Users

WE TEST IT

- Find Wrong Answers
- Trust
- Ease of Use
- Mental Effort

WHAT WE LEARN

How answer type affects users

Previous Datasets

We reviewed the recent studies to understand how datasets are built and used in Bangladeshi legal AI systems.

LawRAG

59,741
Bangladeshi legal documents
(metadata : title, Act number, law family, year

20
manually reviewed gold-answer queries

Large legal corpus with metadata and legal-specific evaluation.

JuristRAG

3 Acts
(Penal Code, CrPC, Special Powers Act)

1,112 sections
(330 pages)

700
curated Q&A pairs

Section-level legal retrieval with a large QA dataset.

LegalRAG

13
Bangladeshi Police Gazettes
(81 pages)

168
Q&A pairs
(factual, temporal, dialect, spelling errors, out-of-context)

Bilingual dataset with diverse and noisy queries.

JusticeNetBD

Women-related laws + NGO / legal-aid information
(paragraph-level chunks)

10
question legal QA benchmark

Evidence-grounded QA for women's legal rights.

Mina

Bangladeshi legal corpus (organized by Acts and Sections)

2022 & 2023
Bangladeshi Bar Council exams
(MCQ + Written + Viva)

Realistic legal reasoning using actual examination questions.

Dataset Requirements

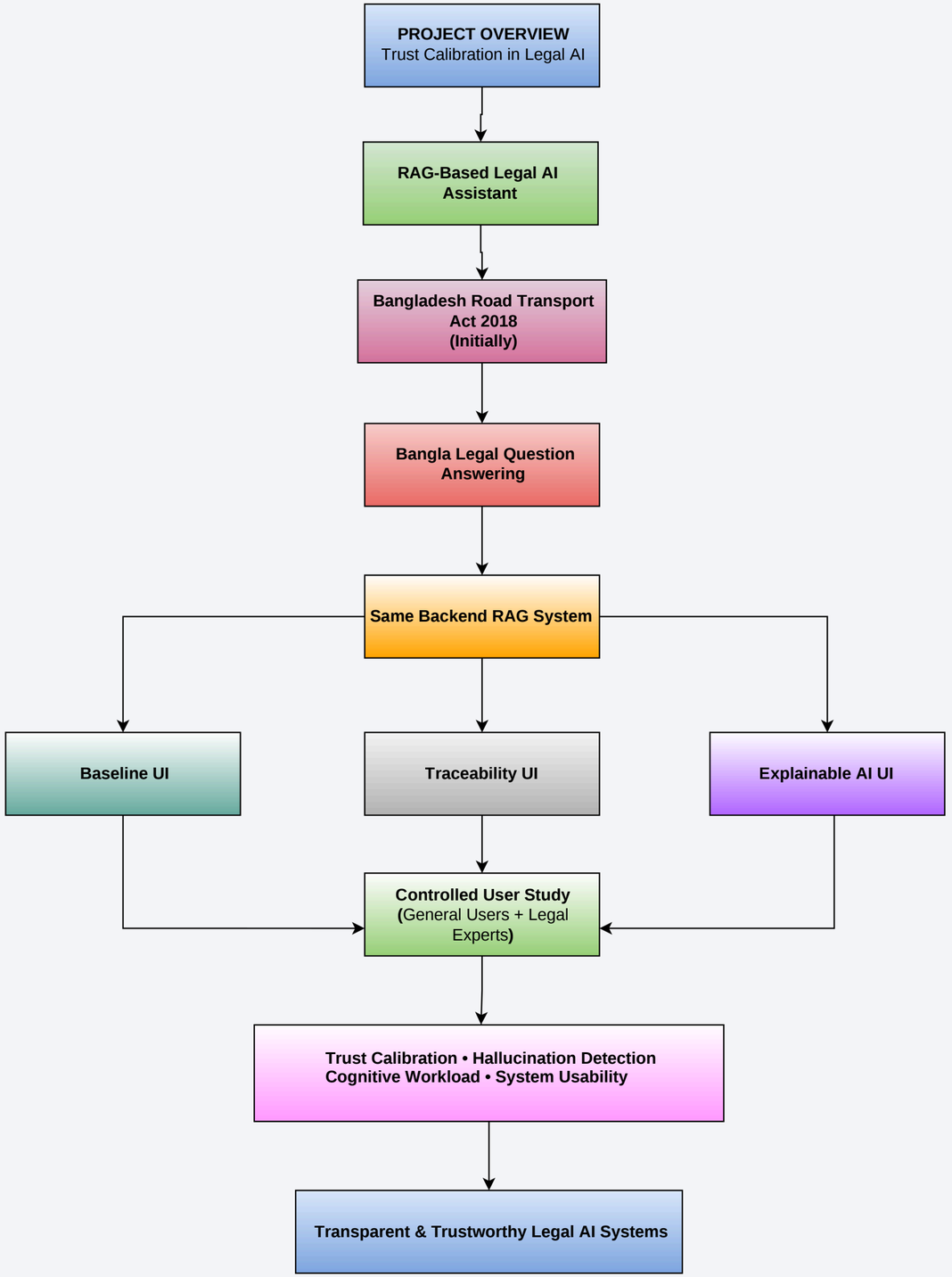
Structured Legal Source

- Act → Section → Subsection
- Original statutory text
- Metadata: title, year, language

Diverse Question Design

- Fact, procedure, penalty
- Scenario and temporal queries
- Bangla + English, noisy queries

- Ground-truth answer + evidence
- Compare with model response
- Groundedness / hallucination label


PROJECT OVERVIEW
Trust Calibration in Legal AI
RAG-Based Legal AI Assistant
Bangladesh Road Transport Act 2018 (Initially)
Bangla Legal Question Answering
Same Backend RAG System
Baseline UI
Traceability UI
Explainable AI UI
Controlled User Study (General Users + Legal Experts)
Trust Calibration • Hallucination Detection
Cognitive Workload • System Usability
Transparent & Trustworthy Legal AI Systems

Project Objective

Main Goal

To create a simple, user-friendly planner that helps students manage deadlines, reduce stress, and stay focused during exam periods.

Supporting Objectives

- Design an intuitive interface suitable for mobile and desktop use
- Include task tracking, reminders, and time-blocking features
- Keep the system lightweight, offline-friendly, and distraction-free



What We Found

1

2/3 of students we surveyed said they feel “overwhelmed” during finals.

2

1-on-1 interviews revealed students want a system that feels personal, not overwhelming.

3

Most rely on scattered notes or mobile reminders, which often get ignored.

Key Insight

Students don’t lack tools — they lack tools that work the way they think.

Who We Designed For



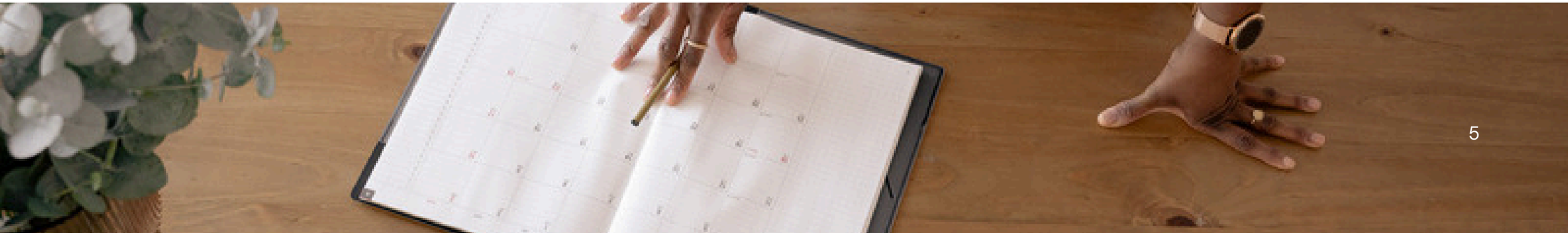
Target User

College students between 18–24 years old, balancing academics, part-time jobs, and personal life.



Their Challenges

- Struggle with managing time and deadlines
- Easily distracted by phones or unstructured routines
- Feel overwhelmed by traditional productivity apps



COST, TIME, AND LAWYER ACCESSIBILITY KEEP EVERYDAY PEOPLE FROM FORMAL LEGAL HELP — SO LLM-BASED ASSISTANTS, IN BANGLADESH AND GLOBALLY, ARE STEPPING INTO THAT GAP FAST.

Cost

Narrowing the Scope

We started with too many ideas and had to make hard decisions about what to keep and what to let go. Prioritizing the most valuable features helped us stay focused.

Challenge 2

User Testing Feedback

Some feedback conflicted with our original plan. It was challenging to adjust without losing our core concept, but we chose the suggestions that best supported our goals.

Problem

Brainstorm → User Feedback → Concept Sketches → Final Idea

Our initial concept was a simple planner app to help students track assignments.

- Feedback showed students felt overwhelmed by typical apps
- We explored visual and tactile planning tools
- Inspiration from bullet journaling, habit tracking, and minimal design

A flexible digital planner that uses visual blocks and subtle reminders to help students stay on track without pressure.

Proposed Dataset Design

Source Layer

- Official statutory law
- Act → Section → Subsection
- Original text + metadata

Authoritative legal source

Question Layer

- Bangla + English questions
- Fact, procedure, penalty
- Scenario, temporal, noisy queries

Diverse query design

Evaluation Layer

- Ground-truth answer + evidence
- Compare with LLM response
- Supported / Partial / Unsupported / Contradicted

Groundedness and hallucination analysis

Methodology

Research

We conducted surveys and informal interviews to understand student pain points.

Ideation

Brainstorming sessions with moodboards and concept sketches helped us narrow our focus.

Prototyping

We created wireframes and mockups to visualize the user flow and core features.

Feedback & Revisions

We gathered peer feedback and refined the layout to improve clarity and usability.

Final Build

What We Learned

Personal Skills Gained

We learned how to manage time effectively, communicate ideas clearly, and stay organized under pressure.

Team Collaboration Lessons

We realized the importance of dividing tasks based on each member’s strengths and checking in regularly to avoid misunderstandings.

Problem-Solving Insights

Through user feedback and iteration, we learned that simple solutions often work better than complex ones. Listening actively was key.

Adaptability

When some ideas didn’t work as expected, we learned how to pivot quickly and stay focused on the end goal.

STUDIO SHODWE

The End

TEAM A

Drew Feig
Eleanor Fitzgerald
Estelle Darcy
Donna Stroupe

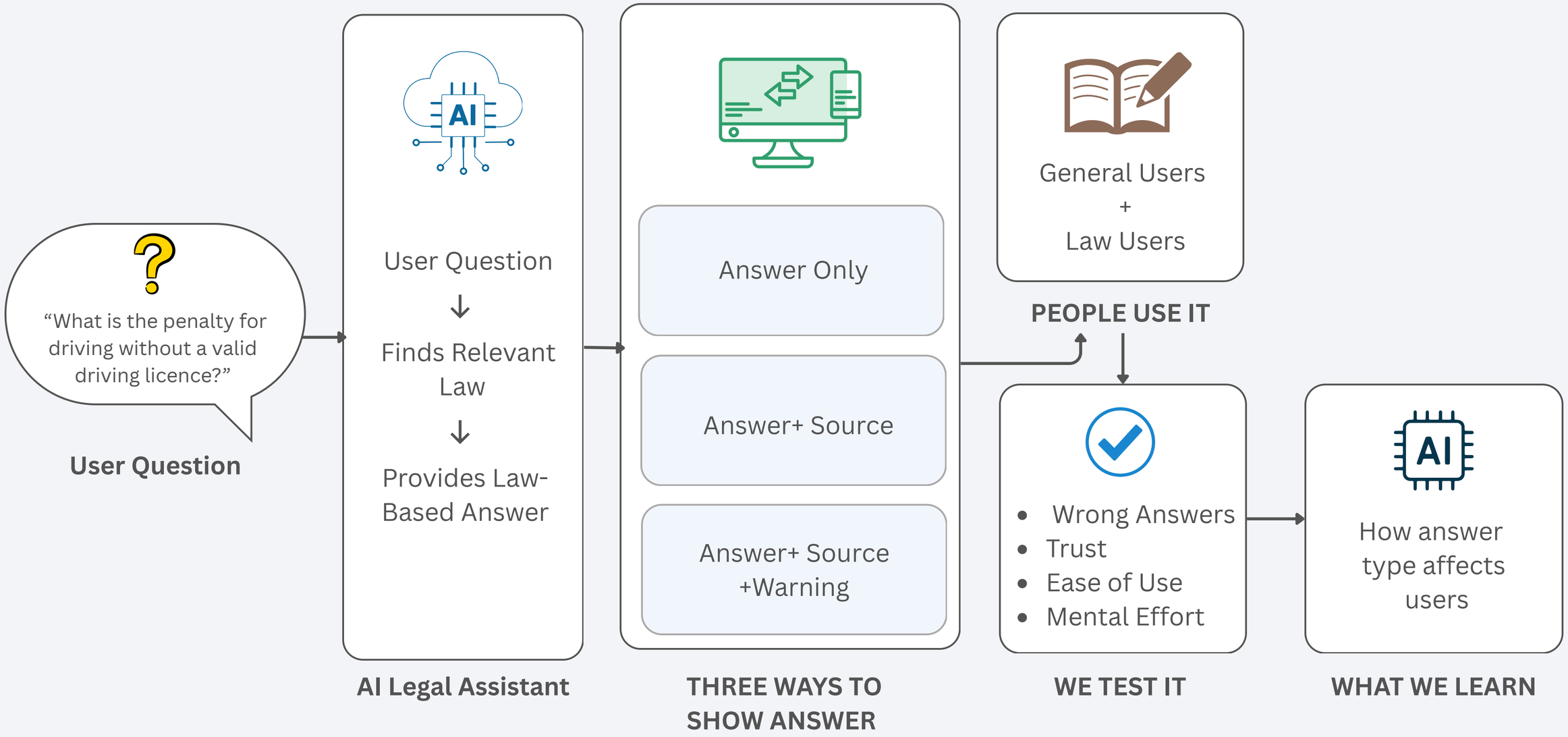
How can we address the difficulty users face in recognizing hallucinated legal information generated by AI?



Gap Area	Existing Work	Our Approach
LEGAL DOMAIN	**GENERAL LEGAL QA:** Broad law coverage	**TRAFFIC-LAW FOCUS:** Bangladesh Road Transport Act
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USER GROUPS	**LIMITED COMPARISON:** Expertise often not studied	**EXPERTISE-AWARE:** Lay users vs. legal experts
DATASET	**LIMITED RESOURCE:** Native-Bangla traffic-law QA	**VALIDATED DATASET:** Expert-reviewed Bangla QA

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EVALUATION	**SYSTEM-CENTRIC:** Accuracy and retrieval metrics ✗	**HUMAN-CENTRIC:** Hallucination detection and trust ✓
TRANSPARENCY	**LIMITED COMPARISON:** Few UI-level studies ✗	**3-INTERFACE STUDY:** Baseline, Traceability, Explainable ✓
USER GROUPS	**LIMITED COMPARISON:** Expertise often not studied ✗	EXPERTISE-AWARE: Lay users vs. legal experts ✓
DATASET	**LIMITED RESOURCE:** Native-Bangla traffic-law QA ✗	**VALIDATED DATASET:** Expert-reviewed Bangla QA ✓

Graphical Abstract



Research Gap

Existing Research

✓ Legal Retrieval
✓ Multilingual RAG
✓ Query Refinement
✓ Hybrid / Iterative Retrieval
✓ System & Expert Evaluation

Remaining Gap

⚠ AI Hallucinations
⚠ Citation / Source Reliability
⚠ User Error Detection
⚠ Trust Calibration
⚠ Interface Effectiveness

Technical Performance → User Response

Can users detect and appropriately respond to incorrect legal-AI answers?